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To:

The Editor-in-Chief

Microelectronics Journal

Elsevier B.V.

Subject: Submission of original research manuscript for publication

Dear Editor,

I am pleased to submit our original research manuscript entitled **"An Ultra Low Power 10 Transistor 1-Bit Magnitude Comparator Using Gate Diffusion Input Technique"** for exclusive consideration for publication as a research article in the *Microelectronics Journal*.

Background and Key Contributions:

Low-power digital logic design is critical for emerging IoT nodes, ultra-portable electronics, and implantable medical devices. In this paper, we present a novel 10-transistor 1-bit magnitude comparator utilizing the Gate Diffusion Input (GDI) technique. Compared to traditional CMOS and pass-transistor logic architectures, our proposed design achieves:

- A significant reduction in transistor count down to 10 transistors, maintaining full swing operation without additional buffers.
- Ultra-low power dissipation and optimized Power-Delay Product (PDP) across varying supply voltages and operational frequencies.
- Superior immunity to process, voltage, and temperature (PVT) variations, making it ideal for energy-harvesting and low-power VLSI applications.

Relevance to the Journal:

Given the *Microelectronics Journal's* focus on advanced microelectronic circuit design, low-power VLSI architectures, and innovative semiconductor design methodologies, we believe our findings will resonate strongly with your readership and contribute meaningfully to low-power digital circuit research.

Declarations:

- **Originality:** This manuscript represents original work and has not been published previously, nor is it currently under consideration for publication elsewhere.
- **Authorship Approval:** All co-authors (**Shahrear Hossain Shawon** and **Riazul Islam**) have read, reviewed, and approved the final manuscript for submission.
- **Conflict of Interest:** The authors declare no conflict of interest regarding the publication of this work.

Thank you very much for considering our manuscript. We look forward to receiving the reviewers' comments in due course.

Sincerely,

Shahrear Hossain Shawon

On behalf of all authors

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